

QUESTION 1 (9 Marks)

- | | Marks |
|---|--------------|
| (a) In a set of 7 letters, some of the letters are T 's and all other letters are different. If the number of different arrangements of these letters is 210, how many letters are T 's. | 2 |
|
 | |
| (b) In a colony of bacteria, the rate of change of the colony is given by:
$\frac{dP}{dt} = kP - r,$ where P is the number of bacteria at time t minutes, r is the constant rate per minute at which the bacteria die and k is a constant. | |
| (i) Verify that $P = \frac{r}{k} - \frac{A}{k}e^{kt}$ is the solution to the rate equation
$\frac{dP}{dt} = kP - r,$ given A is a constant. | 2 |
|
 | |
| (ii) Find the time when the population of the bacteria colony is reduced to zero, given that when $t = 0$, $P = 5000$, $k = 0.2$ and $r = 1500$. Give your answer to the nearest second. | 3 |
|
 | |
| (iii) Find P when $t = 2$, (answer to the nearest bacteria). | 2 |

QUESTION 2 (9 Marks) START A NEW PAGE

- | | Marks |
|--|--------------|
| (a) The velocity $v \text{ cm s}^{-1}$ of a particle is given by $v = 2x + 5$. If the initial displacement is 1cm to the right of the origin, find the displacement as a function of time. | 3 |
|
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| (b) (i) A Brine solution contains 1kg of salt per 10 litres. It runs into a tank, initially filled with 500 litres of fresh water, at a rate of 25 litres per minute. At the same time, the mixture runs out of the tank at the same rate. If A kg is the amount of salt in the tank at time t minutes, Explain why: $\frac{dA}{dt} = 2.5 - \frac{A}{20}.$ | 2 |
|
 | |
| (ii) Find the amount of salt in the tank at the end of 60 minutes, assuming the mixture is kept homogenous (to the nearest 10 grams). | 3 |
|
 | |
| (iii) Find the maximum concentration of salt in the mixture. | 1 |

QUESTION 3 (9 Marks)**START A NEW PAGE**

- | | Marks |
|--|--------------|
| (a) Sixteen of the chickens on the James Ruse School Farm are separated at random into 4 pens of 4 chickens for a feed trial.
What is the probability that 4 particular chickens, A , B , C and D are in 4 separate pens? | 3 |
|
(b) The velocity of a body, $v \text{ ms}^{-1}$, moving in a straight line is given as $v = e^t - e^{-t}$, where t is the time in seconds.
The initial position of the body is at the origin. | |
| (i) Find the displacement x as a function of time t . | 2 |
| (ii) Find the acceleration when $t = 2$.
Give your answer correct to 2 decimal places. | 2 |
| (iii) Show that the body does not have a zero acceleration. | 2 |

QUESTION 4 (9 Marks)**START A NEW PAGE**

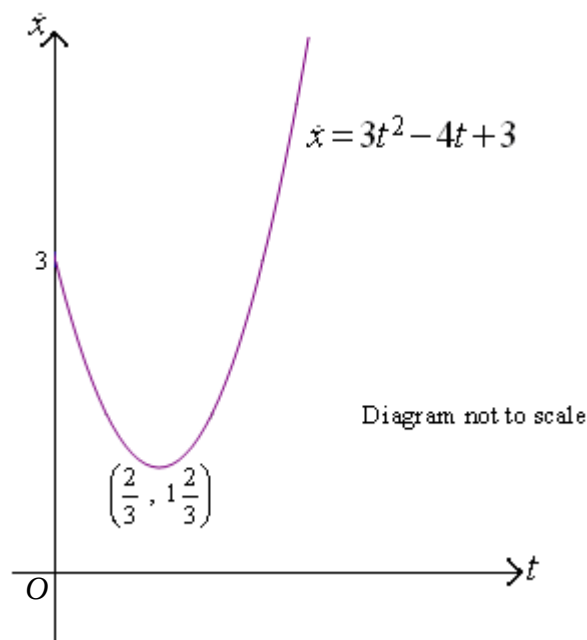
- | | Marks |
|---|--------------|
| The depth of water in y metres on a tidal creek is given by:
$4 \frac{d^2 y}{dt^2} = 5 - y$, where time t is measured in hours. | |
| (i) Prove that the vertical motion of the water level is simple harmonic and hence find the centre of motion. | 2 |
| (ii) Find the period of the motion. | 1 |
| (iii) Given that $y = 2$ at low tide and $y = 8$ at high tide, and that $y = a + b \cos nt$ is the solution of the equation: $4\ddot{y} = 5 - y$, write down the values of a , b and n . | 3 |
| (iv) If the low tide is at 10 am, what is the earliest time after low tide that a fishing boat requiring a depth of 4 metres of water can enter the creek? | 3 |

QUESTION 5 (9 Marks)**START A NEW PAGE****Marks**

(a) Calculate the number of arrangements of the letters *DESCARTES*:

- | | | |
|-------|---|----------|
| (i) | If the two <i>S</i> 's are adjacent. | 1 |
| (ii) | If no two vowels are together. | 2 |
| (iii) | If the conditions from part (i) and (ii) hold simultaneously. | 2 |

(b) The graph below illustrates the velocity of a particle as a function of time.



- | | | |
|------|--|----------|
| (i) | Sketch the graph of the particle to illustrate the acceleration as a function of time, given that the particle is initially 1 m to the left of the origin <i>O</i> . | 2 |
| (ii) | Hence write a description of the motion. | 2 |

QUESTION 6 (9 Marks)**START A NEW PAGE****Marks**

- (a) The velocity $v \text{ ms}^{-1}$ of a particle moving along the x -axis is given by:
 $v = \sqrt{2 + 2 \cos 2x}$. Initially the particle is located at the origin.

(i) Find the initial velocity and acceleration. **3**

(ii) Assuming that the particle reaches the position of $\frac{\pi}{2}$ metres from the origin, determine what would happen to the particle after this time. **2**

- (b) In a certain experiment recording the number of bees N pollinating flowers in a given area, it was found that the rate of change of N is

given by: $\frac{dN}{dt} = kN \left(1 - \frac{N}{2000} \right),$

where t is the time in days and k is a constant.

At the beginning of the experiment 1000 bees were introduced to the area.

(i) Verify that $N = \frac{2000}{1 + e^{-kt}}$ is the solution of the equation. **2**

(ii) If $N = 1500$ when $t = 10$, determine the time in days, when $N = 1800$. **2**

QUESTION 7 (9 Marks)**START A NEW PAGE****Marks**

- (a) A shell is detonated on level ground throwing fragments with a speed $V \text{ ms}^{-1}$ in all directions.
 After a time T , a fragment hits the ground at a distance M from the shell.

You may assume these parametric equations of motion:

$$x = Vt \cos \alpha \text{ and } y = Vt \sin \alpha - \frac{1}{2}gt^2$$

(i) Show that: $g^2T^4 - 4V^2T^2 + 4M^2 = 0$. **2**

(ii) Hence find, to 2 decimal places, the shortest period of time during which a man, standing 20 metres from the place where the shell bursts, is in danger when $V = 25$. Take $g = 10$. **3**

- (b) Twelve politicians are seated at a round table. A committee of five is to be chosen. If each politician, for one reason or another, dislikes their immediate neighbours and refuses to serve on a committee with them, in how many ways can a compatible group of five politicians be chosen? **4**

END OF EXAMINATION

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Suggested Solutions	Marks	Marker's Comments
a) $v = \frac{dx}{dt} = 2x+5$ $\frac{dx}{2x+5} = \frac{1}{2} \ln(2x+5) + c$ When $t=0$, $x=1$ $\therefore 0 = \frac{1}{2} \ln 7 + c \quad \therefore c = -\frac{1}{2} \ln 7$ $\therefore t = \frac{1}{2} \ln \left(\frac{2x+5}{7} \right)$ $e^{2t} = \frac{2x+5}{7} \Rightarrow x = \frac{7e^{2t}-5}{2}$	1 1 1	Several people left final answer in form $x = \frac{e^{2t+\ln 7}-5}{2}$ (Half mark deducted)
b) $\frac{dA}{dt} = \text{Rate In} - \text{Rate Out}$ (A is amount of salt in kg) Rate In = $1 \times \frac{25}{10} = 2.5 \text{ kg/min}$ Rate Out = $25 \times \frac{A}{500} = \frac{A}{20} \text{ kg/min.}$ $\therefore \frac{dA}{dt} = 2.5 - \frac{A}{20}$	1	No second mark was given if A_{20} appeared from nowhere.
ii) Solution of this equation is $A = B + Ce^{-t/20} \quad \text{where } B, C \text{ constants}$ When $t=0$, $A=0 \quad \therefore B = -C$ $\therefore A = B(1 - e^{-t/20})$ But $\frac{dA}{dt} = \frac{Be^{-t/20}}{20} = 2.5 - \frac{B}{20} + \frac{Be^{-t/20}}{20}$ $\therefore B = 50$ $\therefore A = 50(1 - e^{-t/20})$ When $t=60$, $A = 50(1 - e^{-3}) = 47.51 \text{ kg}$	1	If gms used and rounded wrong, last mark NOT given. If just a 3rd digit error in kgms only $\frac{1}{2}$ mark deducted.
iii) As $t \rightarrow \infty$, $A \rightarrow 50$ as $e^{-t/20} \rightarrow 0$. Max concentration, never strictly attained is $\frac{50}{500} \text{ kg/l} = 0.1 \text{ kg/l.}$	1	If stopped at $A=50$, then $\frac{1}{2}$ given. Many people, incorrectly, used $\frac{dA}{dt} = 0$ and arrived at correct answer.

MATHEMATICS Extension 1: Question 3

Suggested Solutions	Marks	Marker's Comments
a) (pens identical) Place A, B, C, D in 4 pens (1 way) Choose groups of 3 to go in each pen from remaining 12 $= {}^{12}_3 \times {}^9_3 \times {}^6_3 \times {}^3_3 = 369600$ No. of ways of choosing 4 groups of 4 from 16 (identical groups) $= \frac{{}^{16}_4 \times {}^{12}_4 \times {}^8_4 \times {}^4_4}{4!} = \frac{63063000}{24} = 2627625$ $\therefore P(\text{ABCD separate}) = \frac{369600}{2627625} = \frac{64}{455}$ Alternatively: pens not identical $\frac{4! \times 369600}{63063000} = \frac{64}{455}$ Alternatively $\frac{16}{16} \times \frac{12}{15} \times \frac{8}{14} \times \frac{4}{13} = \frac{64}{455}$ (Hint: $\frac{16}{16}$ (group) $\times \frac{12}{15}$ (different group) $\times \frac{8}{14}$ (diff gp) $\times \frac{4}{13}$ (diff group))	3	① 369600 ① 2627625 ① 4!
b) (i) $y = e^t - e^{-t}$ $x = \int e^t - e^{-t} dt$ $x = e^t + e^{-t} + C$ $x = C \quad t = 0 \quad \therefore C = -2$ $\therefore x = e^t + e^{-t} - 2$ (ii) $a = \frac{dy}{dt} = e^t + e^{-t}$ $t = 0$ $a = 7.524391382$ accel is 7.52 m/s² (2dp) (iii) $a = e^t + e^{-t}$ $e^t > 0 \text{ for all } t$ $e^{-t} > 0 \text{ for all } t$ $\therefore a > 0 \quad \therefore \text{never zero}$	2	① $x = e^t + e^{-t} + C$ ① $C = -2$ ② ① $a = e^t + e^{-t}$ ① 7.52 (value including 2dp) no deduction if no units ② $e^t > 0$ ① $e^{-t} > 0$ ① [insufficient to say $e^t \neq 0$ $e^{-t} \neq 0$] ① only or $e^t > 0$ $e^{-t} > 0$

QUESTION 4: (9 Marks)

(a) (i) $4 \frac{d^2y}{dt^2} = 5 - y$

$\ddot{y} = -\frac{1}{4}(y-5)$ ①

Which is of the form $\ddot{y} = -n^2(x-b)$, hence the motion is SHM.

Centre of motion occurs when $\ddot{y} = 0$ i.e. at $y = 5$ ①

(ii) $n^2 = \frac{1}{4}$

$\therefore n = \frac{1}{2}$ (taking the positive value)

Period = $\frac{2\pi}{n} = 4\pi$ hours or 12 hrs 34 min. ①

(iii) $y = a + b \cos nt$ a is the centre of motion i.e. $a = 5$ ①

b is the amplitude $\frac{8-2}{2} = 3$

$n = \frac{1}{2}$ from (ii) ①

$\therefore b = -3$ negative sign measures time from low tide ①

$\therefore y = 5 - 3 \cos \frac{1}{2}t$

$\therefore a = 5, b = -3, n = \frac{1}{2}$

(iv) Low tide occurs at 10am.

Fishing boat needs 4m of water i.e. $y = 4$

$\therefore 4 = 5 - 3 \cos \frac{1}{2}t$ ①

$\frac{1}{3} = \cos \frac{1}{2}t$

$t = 2$ hrs 28 minutes

$\therefore$ Required time from low tide is 2 hrs 28 minutes and the actual time is 12:28pm. ①

5a) (i) 2 'e' as 1 unit
8 units with 2 'e' can be arranged in $\frac{8!}{2!} = 20160$ #

in (ii) 6 consonants with 2 's' can be arranged in $\frac{6!}{2!}$ ways
3 vowels A, E, E can occupy 3C_3 positions
Vowels can be ordered in $\frac{3!}{2!} = 3$ ways

$$\text{No. of arrangements} = \frac{6!}{2!} \times {}^3C_3 \times 3 = 37800 \#$$

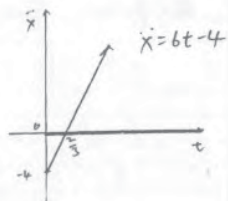
Alternatively $\uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$ 6 consonants 7 gaps
1st vowel has 7 positions
2nd " " 6 " "
3rd " " 5 " "

$$\text{No. of arrangements} = \frac{6!}{2!} \times 7 \times 6 \times 5 = 37800 \#$$

2m

(iii) D C R T (SS) can be ordered in $5!$ ways
3 vowels can occupy 3C_3 positions
Vowels can be ordered in 3 ways
No. of arrangements = $5! \times {}^3C_3 \times 3 = 7200$
Alternatively $5! \times 6 = 7200$
 $2 \leftarrow 2'e'$

b) 2m



(ii) 2m

Initially located 1m left of origin moving to the right ($v > 0$) $\frac{1}{2}m$
slowing down ($v > 0, a < 0$) $\frac{1}{2}m$
min velocity at $t = \frac{2}{3}s$ $\frac{1}{2}m$
Acceleration (force) changes direction to be positive
speeding up to the right ($v > 0, a > 0$) $\frac{1}{2}m$

1m

Any 2 of
 $\frac{6!}{2!} \times {}^3C_3 \times 2$
get 1m
(with product evaluated)

$$\frac{6!}{2!} \times 7 \times 6 \times 5 = 37800 \text{ get 1m}$$

$$5! = 6 \times 5 \times 4 = 120 \text{ get 1m}$$

No graph but
show $x = 6t - 4$ $\frac{1}{2}$
 $\frac{2}{3}s$ for real intercept
straight line 1m

many students started
the straight line to
left of $\ddot{x}$ axis
lose $\frac{2}{3}$ ($t = 0$ only)

bii) many students
did not explain
why particle
move to right $v > 0$
slow down ($v > 0, a < 0$)
speed up ($v > 0, a > 0$)

Complimentary Method

5ii) 2 'e' eea
eae
aee
1 'e' 1 'a' eae eae
eea
aee

No of Restrictions - ($ae + ie'la'$) + ($eea + aee + eae$)

$$\frac{9!}{2!2!} - \left[\frac{8!}{2!} + \frac{8!}{2!} \times 2 \right] + \left(\frac{7!}{2!} \times 3 \right) \quad 1m$$

$\uparrow$ 2e
 $\uparrow$ a, e
can swap

$$= 90720 - [20160 + 40320] + 7560$$

$$= 37800$$

1m

MATHEMATICS Extension 1: Question 6		Marks	Marker's Comments
Suggested Solutions			
b (a) $v = \sqrt{2+2\cos 2x}$ (i) $a = \frac{dv}{dx} \left(\frac{1}{2} v^2 \right)$ $= \frac{d}{dx} \left(\frac{1}{2} (2+2\cos 2x) \right)$ $= \frac{d}{dx} (1+\cos 2x)$ $= -2\sin 2x$ OR $a = v \cdot \frac{dv}{dx}$ $= \sqrt{2+2\cos 2x} \cdot \frac{1}{2} (2+2\cos 2x)^{-\frac{1}{2}} \cdot -4\sin 2x$ $= -2\sin 2x$ When $t=0$, $x=0 \Rightarrow v = \sqrt{2+2\cos 0} = 2\text{ m/s}$ $\Rightarrow a = -2\sin 0 = 0\text{ m/s}^2$	1	1	** For (a)(ii) See over page [3]
(b)(i) If $N = \frac{2000}{1+e^{-kt}}$ show $\frac{dN}{dt} = kN \left(1 - \frac{N}{2000} \right)$ Method 1: $LHS = 2000(-1)(1+e^{-kt})^{-2} \cdot -ke^{-kt}$ $= \frac{2000ke^{-kt}}{(1+e^{-kt})^2}$ $RHS = kN \left(1 + \frac{N}{2000} \right)$ $= k \times \frac{2000}{1+e^{-kt}} \left(1 - \frac{1}{1+e^{-kt}} \right)$ where $\frac{N}{2000} = \frac{1}{1+e^{-kt}}$ $= \frac{2000k}{1+e^{-kt}} \left(\frac{1+e^{-kt}-1}{1+e^{-kt}} \right)$ $= \frac{2000ke^{-kt}}{(1+e^{-kt})^2}$ $\therefore LHS = RHS$ as required	1	1	[2]
Method 2: $LHS = \frac{2000ke^{-kt}}{(1+e^{-kt})^2}$ $= k \times \frac{2000}{1+e^{-kt}} \cdot \frac{e^{-kt}}{1+e^{-kt}}$ $= kN \left(\frac{1+e^{-kt}-1}{1+e^{-kt}} \right)$ $= kN \left(1 - \frac{1}{1+e^{-kt}} \right)$ but $\frac{N}{2000} = \frac{1}{1+e^{-kt}}$ $= kN \left(1 - \frac{N}{2000} \right)$ $= RHS$ as required			

MATHEMATICS: Question 6 Extension 1		Marks	Marker's Comments
Suggested Solutions			
Method 3: $LHS = RN \frac{e^{-kt}}{1+e^{-kt}}$ $= kN \left(\frac{\frac{2000-N}{N}}{1+\frac{2000-N}{N}} \right)$ $= RN \left(\frac{2000-N}{N+2000-N} \right)$ $= RN \left(\frac{2000-N}{2000} \right)$ $= RN \left(1 - \frac{N}{2000} \right)$ $= RHS$ as required	1	1	
b(ii) $N=1500$, $t=10 \Rightarrow 1500 = \frac{2000}{1+e^{-10k}}$ Find k $1+e^{-10k} = \frac{4}{3}$ $e^{-10k} = \frac{1}{3}$ $-10k = \ln \frac{1}{3}$ $k = -\frac{1}{10} \ln \frac{1}{3}$ OR $k = \frac{1}{10} \ln 3$	1	1	
When $N=1800$, find t : $1800 = \frac{2000}{1+e^{-kt}}$ $1+e^{-kt} = \frac{10}{9}$ $e^{-kt} = \frac{1}{9}$ $t = -\frac{1}{k} \ln \frac{1}{9}$ $\therefore t = \frac{1}{k} \ln 3^2$ $= \frac{2}{k} \ln 3$ $= \frac{2 \times 10}{\ln 3} \times \ln 3$ $t = 20 \text{ hours}$ $t = \frac{5}{6} \text{ days}$	1	1	
** (a)(ii) at $x = \frac{\pi}{2}$, $v = \sqrt{2+2\cos \pi} = 0$ $a = -2\sin \pi = 0$ $\therefore$ When particle reaches $x = \frac{\pi}{2}$ m, the particle will come to rest and remain there, as there is no force acting on the particle	1	1	-1 if left in wrong units. [2] Need to say why the particle remains at rest, otherwise -1/2 penalty.

* QUESTION 7: (9 Marks)

(a) (i) Initial speed $V \text{ ms}^{-1}$ at an angle of α°

$$x = vt \cos \alpha$$

$$t = T, x = m, y = 0$$

$$m = VT \cos \alpha$$

$$\cos \alpha = \frac{m}{VT}$$

$$\therefore \sin \alpha = \frac{\sqrt{V^2 T^2 - m^2}}{VT} \quad (1)$$

$$y = vt \sin \alpha - \frac{gt^2}{2}$$

$$0 = VT \sin \alpha - \frac{gT^2}{2}$$

$$0 = VT \cdot \frac{\sqrt{V^2 T^2 - m^2}}{VT} - \frac{gT^2}{2}$$

$$\sqrt{V^2 T^2 - m^2} = \frac{gT^2}{2}$$

$$V^2 T^2 - m^2 = \frac{g^2 T^4}{4}$$

$$\therefore g^2 T^4 - 4V^2 T^2 + 4m^2 = 0 \quad (1)$$

(ii) when $m = 20 \text{ m}$ $V = 25 \text{ ms}^{-1}$ $g = 10 \text{ ms}^{-2}$

$$\therefore 100T^4 - 4 \times (25)^2 + 4 \times (20)^2 = 0$$

$$100T^4 - 2500T^2 + 1600 = 0$$

$$T^4 - 25T^2 + 16 = 0 \quad (1)$$

$$T^2 = \frac{25 \pm \sqrt{25^2 - 4 \times 1 \times 16}}{2}$$

$$T^2 = \frac{25 \pm \sqrt{561}}{2}$$

$$T = 4.934, 0.811 \quad (1)$$

$\therefore$ Shortest period of time is 0.8 sec . (nearest sec.)

(b) Case 1

If a particular politician, Mr X, is to be on the committee. The 2 people sitting next to Mr X will not be on the committee.

Therefore, there are 4 to choose from 9. (1)

Let C represent politicians who will be on the committee and C' those who will not be on the committee.

$$\uparrow C' \uparrow C' \uparrow C' \uparrow C' \uparrow C' \uparrow$$

(1) There are 6 places available for those on the committee. i.e. 6C_4 ways of choosing the committee with Mr X on it.

Case 2:

If Mr X is not on the committee there are 7C_5 ways of choosing. (1)

$$\therefore \text{The total no. of ways} = {}^7C_5 + {}^6C_4 = 36 \quad (1)$$

7(b) (Method 2)

0 0 0 0 0 0 0 0
1 2 3 4 5 6 7 8 9



1 3 5 8 9

1 3 5 8

1 3 5 9

1 3 6 8

1 3 6 9

1 3 7 9

1 4 6 8

1 4 6 9

1 4 7 9

1 5 9 9

2 4 6 8

2 4 6 9

2 4 7 9

2 5 7 9

3 5 7 9

Ways to pick first person = 12

Ways to choose these 4 remaining

15 ways.

$$\therefore \text{total} = 15 \times 12 = 180$$

But 1st person interchangeable with other 4 (which are fixed)

So need to divide by

$$\frac{5!}{4!}$$

$$\therefore \text{Answer} = \frac{180}{\frac{5!}{4!}} = \underline{\underline{36}}$$